$\Rightarrow a_j = [b_1 \dots b_p] \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_p \end{bmatrix} \Rightarrow a_j = [b_1 \dots b_p] c_j$

$a_j = Bc_j$ □

b) Let $C = [c_1 \dots c_q]$. Explain why there is a nonzero vector u such that $Cu = 0$.

Soln: By part (a), we know that $\forall j, 1 \le j \le q, c_j \in \mathbb{R}^p$.

$\therefore C$ is a $p \times q$ matrix. We know $q > p$.

We know, the matrix C can have atmost p pivot columns (By condition of echelon form, defn. of pivot and the fact that C can be converted to a unique pref by Row-Reduction Algorithm and Thm 1 from Chapter 1).

$\therefore$ There are $(q-p)$ free variables of C.

By Theorem 3 from Chapter 1, the matrix equation $Cu = 0$ has the same solution set as the system of equations corresponding to the augmented matrix $[C\ 0]$.

By Theorem 2 from Chapter 1, there are infinitely many solutions of the above system of eqns.

$\therefore$ There is a nonzero vector u s.t. $Cu = 0$ □

c) Use B and C to show that $Au = 0$. This shows that the columns of A are linearly dependent.

Soln: By part (b), we know that $Cu = 0$ and $\exists u \neq 0$.

$B \cdot (Cu) = B \cdot 0 \Rightarrow (BC)u = B \cdot 0$ (Theorem 2(a))

$\Rightarrow (B[c_1 \dots c_q])u = 0 \Rightarrow ([Bc_1 \dots Bc_q])u = 0$ (Defn. of matrix multiplication)

$\Rightarrow ([a_1 \dots a_q])u = 0$ (By part (a))

$\Rightarrow Au = 0$. $\therefore$ By defn, the columns of A are linearly dependent. □

28) Use Exercise 27 to show that if A and B are bases for a subspace W of $\mathbb{R}^n$, then A cannot contain more vectors than B, and, conversely, B cannot contain more vectors than A.